

F24-015-D-Sketch2Face

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Chapter 1

Introduction

This document sets forth the specification for Sketch2Face app, an advanced artificial intelligence program aimed at recreating realistic human facial images given a set of sketches or descriptions of faces. It can even manipulate picture according to prompt of user. It is suitable for forensic investigators to convert the sketches to realistic facial images for identification of suspect and even average user can convert their sketches to realistic images with a click of button.

This document is intended for the following types of readers:

1. **Developers:** To understand the system's architecture, APIs, and individual modules that need to be implemented for facial image generation, manipulation, and web integration.
2. **Project Managers:** For managing timelines, coordinating development, tracking milestones, and ensuring the smooth growth of new features and functionality.
3. **Marketing Staff:** To highlight the key features and advantages of the application, promoting it to potential users such as law enforcement agencies, digital artists, and content creators.
4. **End Users:** To navigate the system, generate and modify facial images from sketches or text, and use the tool effectively for forensic investigations or content creation.
5. **Testers:** To perform rigorous testing on the application's features, ensuring that it operates optimally and meets the project's specifications.
6. **Documentation Writers:** To create user manuals, guides, and technical instructions, ensuring that all users have the necessary documentation to operate the system effectively.

1.1 Problem Statement

Creating real human facial images from inputs, like sketches, partial photos, or text, is a complex and unresolved problem. Despite advancements in artificial intelligence, existing solutions (reference 1) fall short in generating high quality real-images that truly transform to human face. These limitations are particularly problematic for law enforcement, where the accurate reconstruction of a face from a sketch can be critical to identify the suspect. This project aims to address this gap by developing an AI-driven system that not only generates realistic human faces from but also offers powerful editing capabilities through a user-friendly application for agencies to modify image according to their needs.

1.2 Scope

Our proposed project aims to develop a AI-driven system capable to convert both hand-drawn sketches and textual prompts into realistic human facial images. This makes are system valuable for forensic investigation to catch suspect where CCTV footages are not available they really on sketches to identify suspect. Moreover artist use to convert their sketches to realistic looking facial images.

One of the key features of our system is manipulating and editing the generated images based on user input. Users can modify specific facial features, such as face shape, hairstyle, hair color, eye color, and more, simply by providing prompts. This flexibility allows users to generate images according to their specific needs and preferences. This helps in forensic investigations.

Moreover, our system supports real-time conversion of sketches into images, allowing users to see the image being created dynamically as they draw. This real-time interaction enhances usability, and helps adjusting during the image creation process.

The boundaries of this project are focused on facial image generation and manipulation. While it helps in creating and editing facial features, it is not extended to full-body generation or other types of images. Moreover, the system will be deployed as a web application, ensuring accessibility to a wide range of users.

1.3 Modules

We have divided our project into three modules. Each module has its own features. Following are the modules that are going to be developed in this project :

1.3.1 Sketch to Facial Images

This module focuses on generating facial images through hand-drawn sketches by utilizing AI models. The model will accurately capture facial details from sketches and convert them into realistic looking facial images.

1. Generate Facial Images through hand-drawn sketches.
2. Generation of High Quality images through complete or partially drawn sketches.

1.3.2 Facial Image Generation and Manipulation Using Text

This module focuses on the model that will generate facial images through textual descriptions provided through prompts. Users will be able to modify facial attributes of the generated image by providing feedback.

1. Generate Facial Images through textual descriptions.
2. Manipulation of Facial Images through textual descriptions provided via prompts.

1.3.3 Web Application

In this module, we will be developing a web application that will allow the user to draw sketches as well as enter textual descriptions via prompts to generate facial images. This module also focuses on developing a user-friendly interface.

1. Interface for drawing sketches and entering textual prompts.
2. Real-time Conversion of Sketches to Facial Images.
3. Integration of AI models with web application.
4. Implementing APIs for communication between AI models and web application.

1.4 User Classes and Characteristics

Identify the various user classes that you anticipate will use this product, and describe their pertinent characteristics.

Table 1.1: User Classes and Characteristics

User Class	Description
Law Enforcement Agencies	Law enforcement agencies will use Sketch2Face to generate realistic facial images from witness sketches during criminal investigations. The system allows for rapid, accurate image generation from sketches or descriptions, significantly improving the identification process and assisting in catching suspects efficiently, reducing investigation time.
Forensic Experts	Forensic experts require a tool that can accurately reconstruct faces from textual descriptions or partial images. Sketch2Face allows them to fine-tune facial features based on witnesses' descriptions, enabling forensic teams to create more accurate images of suspects. This can also be used to enhance or manipulate facial images for case documentation.
Digital Artists	Artists and content creators can leverage Sketch2Face to transform their sketches into realistic facial images. This will be particularly useful in fields like digital content creation, entertainment, or artistic presentations, where generating lifelike facial images quickly and with creative freedom is crucial.
Government Agencies	Government bodies tasked with public safety can use Sketch2Face to generate accurate images from incomplete data, such as during national security investigations or for creating visuals to assist in public warnings. The system can be employed to enhance and edit suspect images or create public safety visuals quickly and accurately.
General Public (Casual Users)	Casual users, including individuals use Sketch2Face to create or manipulate realistic facial images for personal projects, social media content, or fun. The system allows easy access to AI-driven image generation, enabling anyone to create real facial images from simple sketches or descriptions without professional knowledge.
Content Creators	Content creators working on visual storytelling or media projects can use the system to generate realistic characters from sketches. This reduces production time and enhances creativity, allowing them to generate high-quality visuals.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-Case Table

For Sketch2Face, an Use-Case Table is suitable, it involves clear interactions between users and the system.

Table 2.1: Use-Case Table for Sketch2Face

Use Case	Description	Actors	Preconditions	Postconditions
Generate Facial Image from Sketch	User uploads or draws a sketch, and the system generates a realistic facial image.	Law Enforcement, Artists, General Users	A sketch is uploaded or drawn.	A realistic facial image is generated.
Generate Facial Image from Text	User inputs textual descriptions (e.g., "male, short hair, blue eyes") to generate a facial image.	Law Enforcement, Forensic Experts	Text descriptions are provided.	A realistic facial image is generated.
Modify Facial Features Based on Prompts	User modifies specific features of the generated facial image (e.g., change hair color, adjust jawline).	Artists, Forensic Experts, General Users	A facial image is already generated.	The facial image is modified as per the input prompt.
Real-time Sketch Conversion	The system converts a hand-drawn sketch in real time, showing the image as the sketch is being created.	Law Enforcement, Artists	User starts drawing a sketch.	The image is generated dynamically as the user draws.
Save Generated Image	User saves the generated or modified image to their device.	All Users	A facial image is generated.	The image is saved successfully.

2.2 Functional Requirements

This section describes the functional requirements of the system expressed in the natural language style. The requirements are organized by system feature, where each module contains specific functional requirements associated with the feature it supports.

2.2.1 Module 1: Sketch to Facial Images

Following are the requirements for Module 1, which focuses on generating facial images from hand-drawn sketches:

1. **FR1:** The system shall allow users to upload or draw hand-drawn sketches and generate facial images from those sketches using AI models.
2. **FR2:** The system shall generate high-quality facial images based on complete or partially drawn sketches.
3. **FR3:** The AI model shall accurately capture facial features and details from sketches to convert them into realistic facial images.
4. **FR4:** The system shall provide options for users to edit or refine the sketches before generating the final facial image.

2.2.2 Module 2: Facial Image Generation and Manipulation Using Text

Following are the requirements for Module 2, which focuses on generating and manipulating facial images based on textual descriptions:

1. **FR1:** The system shall allow users to input textual descriptions (e.g., "short hair, blue eyes") to generate facial images.
2. **FR2:** The system shall support text-based manipulation of facial images by allowing users to modify specific facial attributes.
3. **FR3:** The system shall provide feedback on changes and display real-time adjustments as users modify the facial image using text inputs.

2.2.3 Module 3: Web Application

Following are the requirements for Module 3, which focuses on the development of a web application for user interaction:

1. **FR1:** The system shall provide a user-friendly interface that allows users to draw sketches and enter textual descriptions to generate facial images.
2. **FR2:** The system shall support real-time conversion of hand-drawn sketches into facial images, dynamically updating the image as the user draws.
3. **FR3:** The system shall integrate AI models for both sketch-to-image and text-to-image generation.
4. **FR4:** The system shall implement APIs for communication between the AI models and the web application to ensure smooth functionality.
5. **FR5:** The system shall provide options to save facial images through the web application.

2.3 Non-Functional Requirements

This section specifies nonfunctional requirements. These quality requirements are specific, quantitative, and verifiable.

2.3.1 Reliability

The system shall ensure a high level of reliability to support continuous facial image generation and manipulation operations.

- **R-1:** In the event of system failure, recovery procedures shall be automatic, and the system should restore services, including facial image generation, within 3 minutes.
- **R-2:** Any critical failure in image generation or manipulation processes must trigger an automatic logging of the incident for future analysis.
- **R-3:** The system shall be able to handle 5 user requests simultaneously, ensuring consistent performance even under high usage scenarios, without degradation in reliability.

2.3.2 Usability

The system shall be user-friendly and easy to navigate for users of all technical levels, including law enforcement officers, artists, and general users.

- **USE-1:** The system shall allow users to generate a facial image from a sketch with no more than three interactions (e.g., upload sketch, draw sketch, and prompt).
- **USE-2:** The system shall provide clear and concise instructions at every step to guide the user through the sketch or text input processes.
- **USE-3:** A user should be able to undo any modifications made to the generated facial image with a single action.

2.3.3 Performance

Performance requirements for various operations are critical to ensure the efficiency of the system.

- **PER-1:** Facial image generated from sketch shall be completed within 15 seconds of submission, given a standard input size.
- **PER-2:** Facial image generated from textual descriptions shall be completed within 20 seconds.
- **PER-3:** The web application interface shall load within 5 seconds for users with a 20 Mbps or faster Internet connection.

2.3.4 Security

The system shall ensure that the facial images generated and any user data are protected from unauthorized access.

- **SEC-1:** The system shall require user authentication before any access to personal data or saved facial images.
- **SEC-2:** Attempts to break into the system shall be logged, and after five consecutive unsuccessful login attempts, the account shall be locked for 15 minutes.

2.4 Domain Model

Below is the domain model for our project Sketch2Face

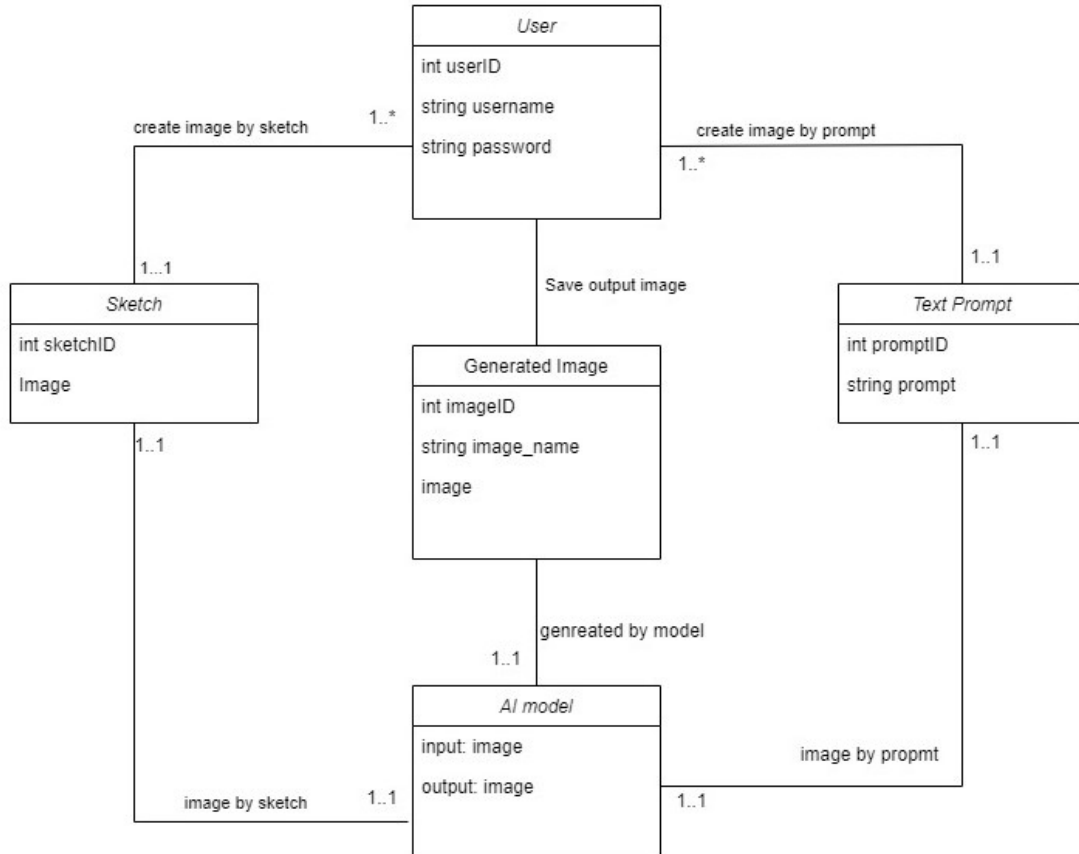


Figure 2.1: Domain Model

Chapter 3

System Overview

This chapter provides the description of the functionalities involved along with the context and design of our proposed system.

3.1 System Workflow

Following is the workflow of our system :

- The user is asked to generate a facial image of a sketch or a textual description.
- Firstly, if the user selects to generate the facial image through a sketch, then he can either upload a sketch or draw by himself.
- The sketch is then denoised and passed to the Sketch-To-Face model which then generates an upsampled image.
- Secondly, if the user enters a prompt, the prompt is converted into tokens and encoded.
- The Text-To-Face Model then generates the facial image.
- Lastly, the user can manipulate the facial attributes of the generated face.

3.2 Architectural Design

The system is based on a Client-Server architecture in which the user interacts with system using a web interface.

3.2.1 Client-Server Architecture

The communication between the web interface and the backend components is done through an API Gateway. The backend consists of three modules which includes Sketch to Face Module, Text to Face Module and Face Manipulation Module. The input is passed through these individual modules. Each module performs a specific task. The output of these modules is stored in a Database.

3.2.2 Client Side

The client is responsible for the communication between the system and the user. It consists of the following two modules :

- **Web Portal :** The user interacts with the system using a Web Portal. This module is responsible for all the front-end related tasks. It consists of the following functionalities :
 - Allow the user to draw or upload Hand-drawn sketches.
 - Allow the user to enter textual prompts for facial image generation.
 - Allow the user to manipulate facial attributes of a generated image.
 - Display the generated output to the user.
- **API Gateway :** This module is used to make communication between the web portal and the backend system possible using API Calls. It is responsible for handling requests and responses. It is used to handle API Clients in an efficient manner.

3.2.3 Server Side

The server is responsible for handling all the backend components such as the Database. It consists of all the services that are required by the system. Following modules are involved in this:

- **Sketch to Face Module :** This module is responsible for the generation of facial images through sketches. The module takes a sketch as an input through API and uses it to generate a similar facial image.
- **Text to Face Module :** This module is responsible for the generation of facial images through textual prompts given by the user. The module takes a textual prompt as an input and then generates a facial image based on the description provided.

- **Sketch to Face Module :** This module is responsible for the manipulation of facial attributes of the generated facial image. The user specifies the facial attributes that needs to be manipulated. The module takes the generated facial image and manipulates the facial attributes as specified by the user.
- **Database :** This module is responsible for storing all the related data that is generated by the system for a specific user.

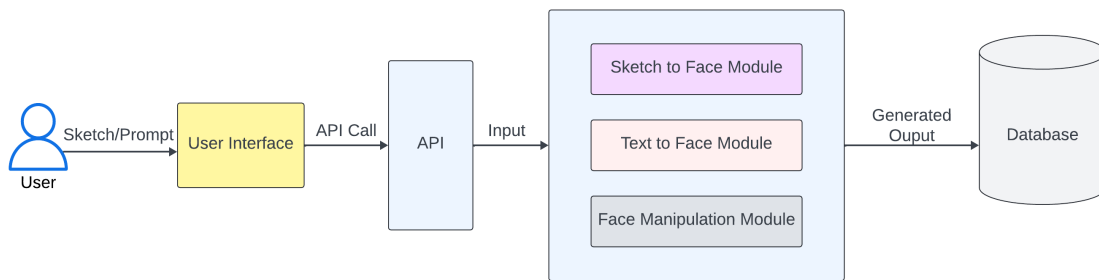


Figure 3.1: Architecture Diagram

3.3 Design Models

The design models for the Sketch2Face system are captured in the **procedural approach**. The reason behind going for the procedural approach is that it offers a simple way of addressing the flow like and linear nature of the Sketch2Face where the user inputs such as sketches or text prompts are processed in a sequential manner to create and transform facial images. The following diagrams assist in understanding how the system is structured and how the different parts of the system are linked and how the system transforms user inputs into final facial image.

3.3.1 Activity Diagram

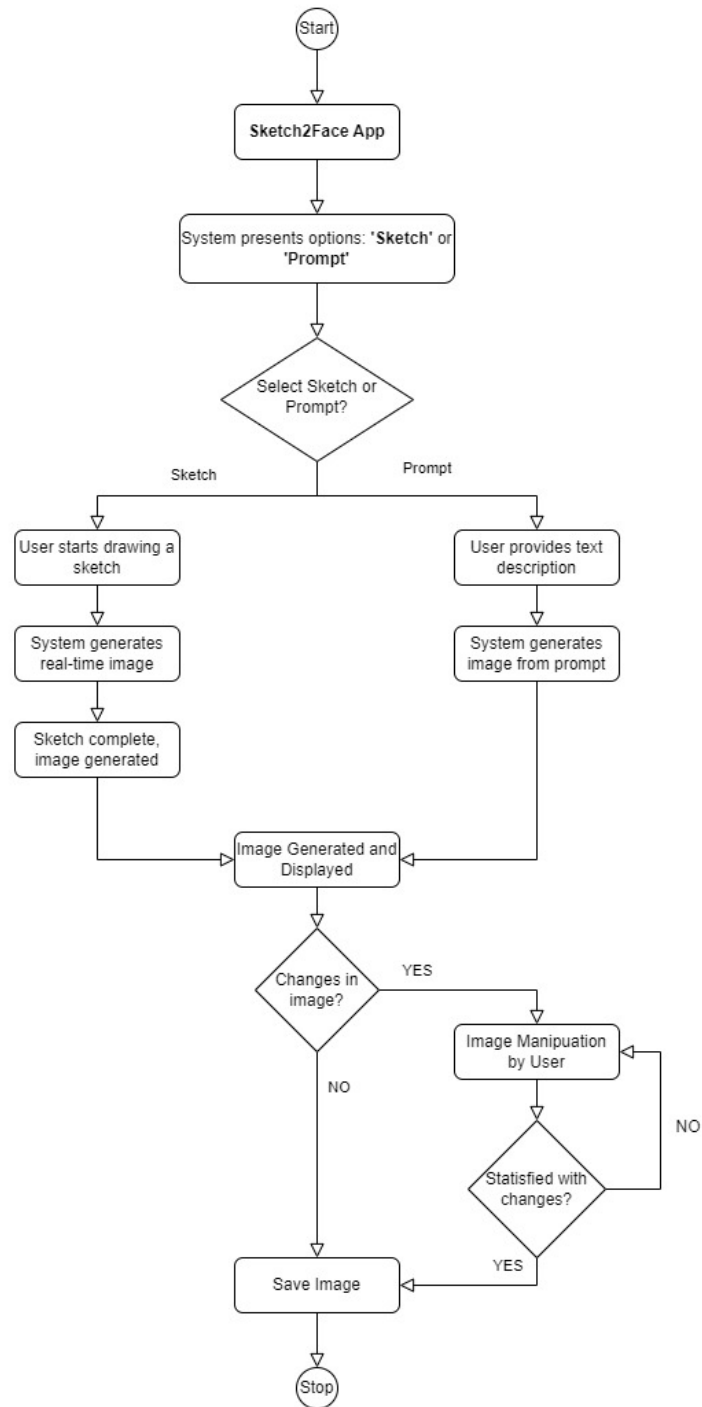


Figure 3.2: Activity Diagram

3.3.2 Data Flow Diagram

The Data Flow Diagram is used to conceptualize the flow of data in the proposed system. It defines the overall design of the system and sub-systems that are implemented. The system is expanded to level 0, 1 and 2 diagrams respectively.

3.3.2.1 Data Flow Diagram Level 0

The level 0 Data Flow Diagram represents the overall functionality of the proposed system. It provides a broader aspect of the services provided by the system. It defines the inputs and the outputs that the system generates associated with each functionality.

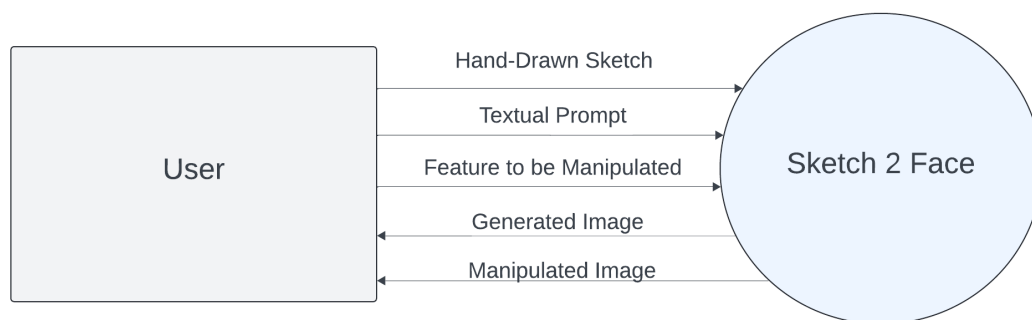


Figure 3.3: DFD Level 0

3.3.2.2 Data Flow Diagram Level 1

The level 1 Data Flow Diagram further explains the sub-systems involved in the system. It describes the flow of data through these individual sub-systems. The "Image Generation 1.0" takes the Hand-drawn sketches and generates the facial image. Whereas "Image Generation 2.0" takes the textual prompt and generates a facial image. "Image Manipulation 3.0" takes the generated facial images and manipulates the facial attributes on the basis of the input provided.

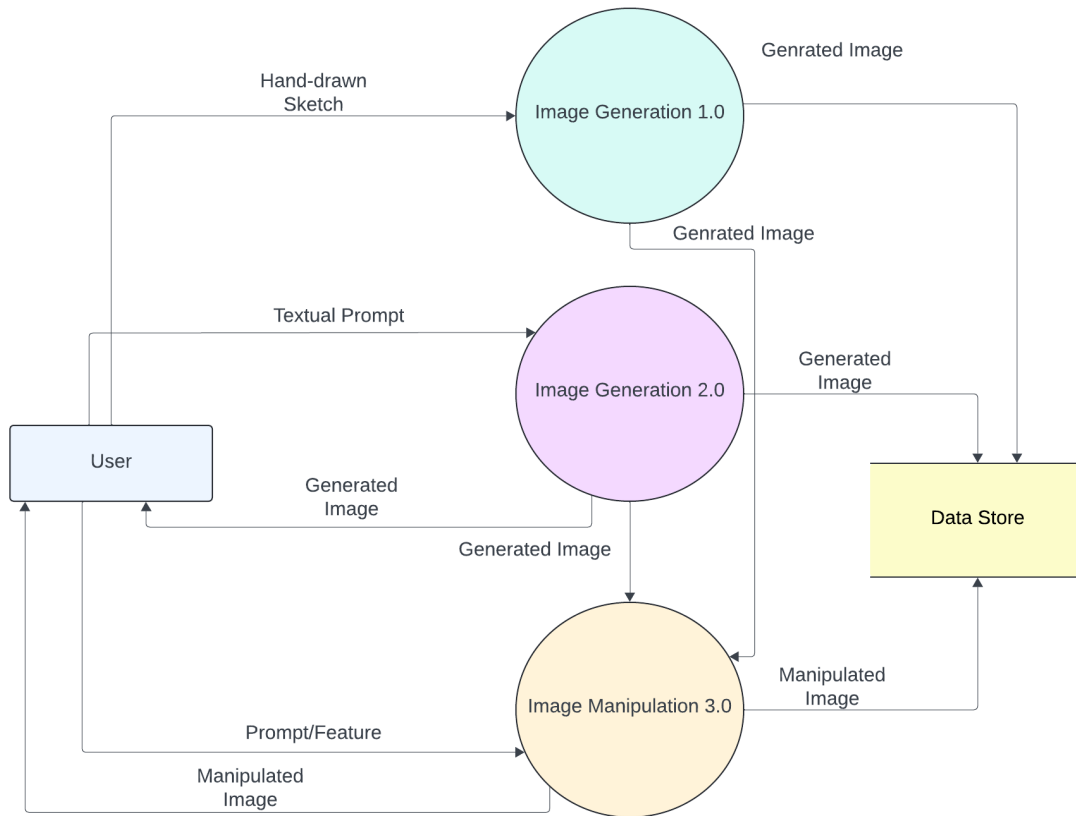


Figure 3.4: DFD Level 1

3.3.2.3 Data Flow Diagram Level 2

- The following Level 2 Data Flow Diagram explains the "**Image Generation 1.0**" sub-system. A hand-drawn sketch is provided which is then passed to the Noise Removal component. The denoised image is then passed to a model whose output is then upsampled and displayed to the user.

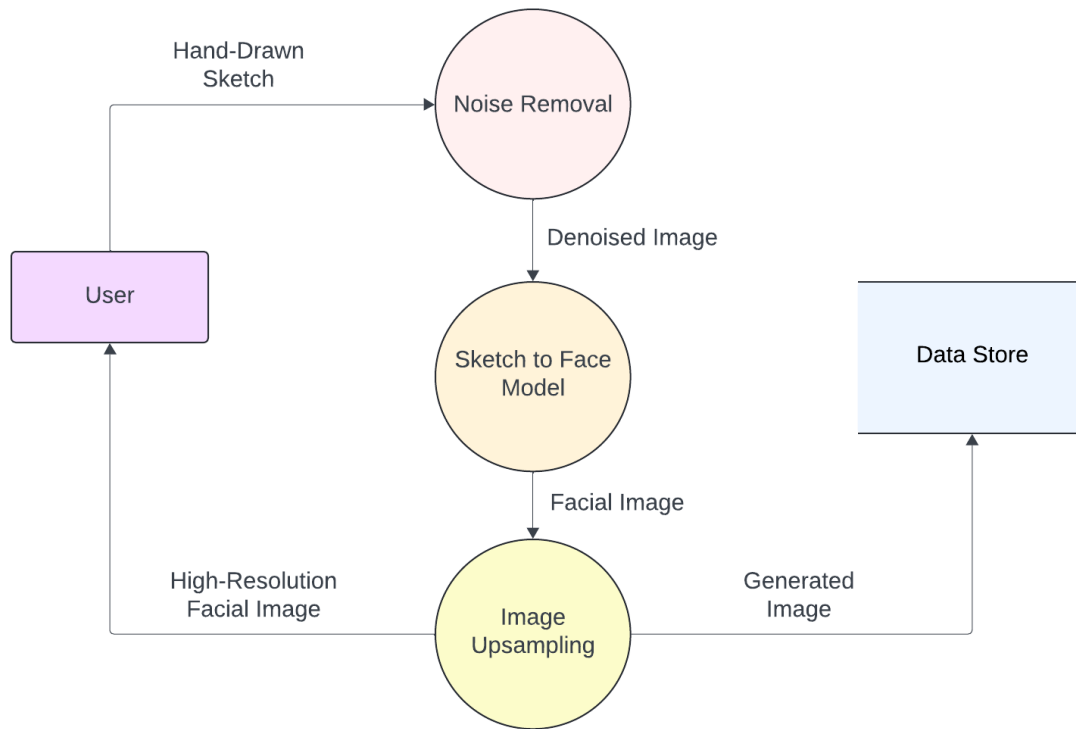


Figure 3.5: Image Generation 1.0

- The Data Flow Level 2 diagram of "**Image Generation 2.0**" takes the textual prompt as an input and is tokenized. After Encoding, a facial image is generated using the text to face model. The generated image is displayed to the user.

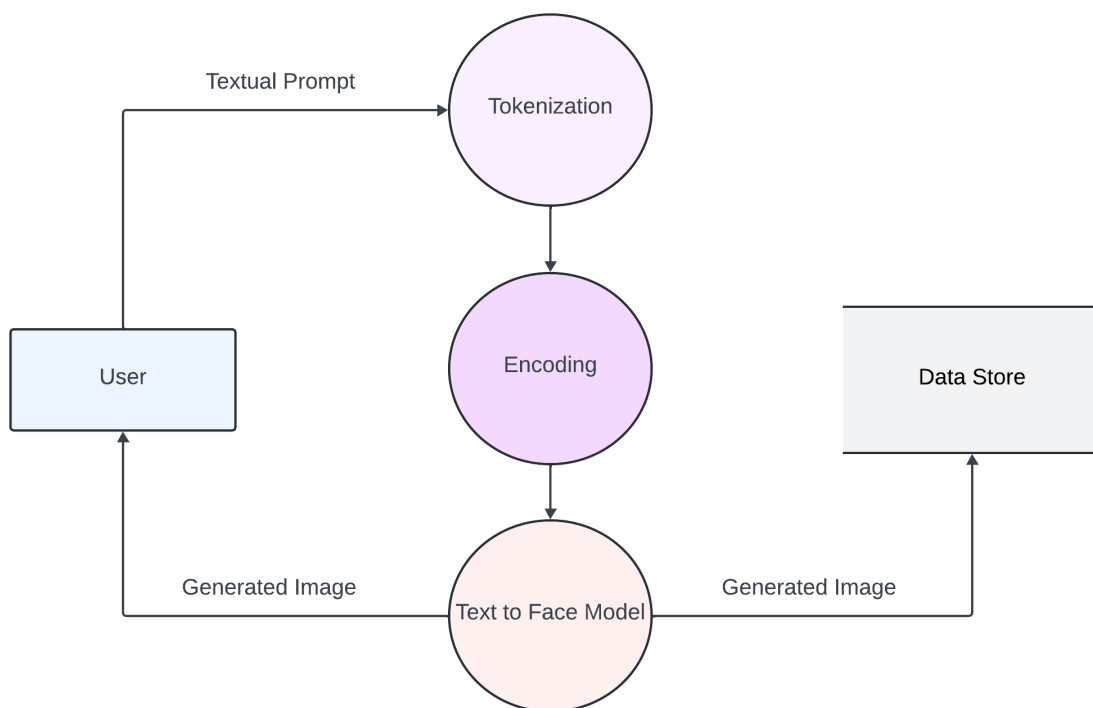


Figure 3.6: Image Generation 2.0

3.3.3 System-Level Sequence Diagram

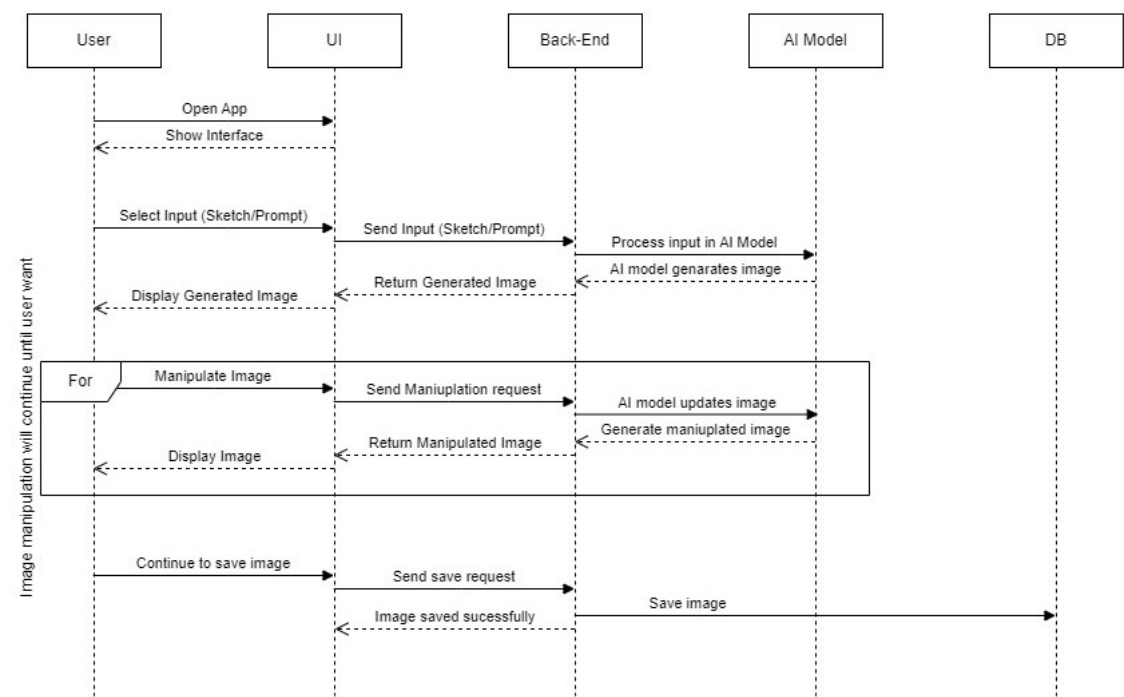


Figure 3.7: System-Level Sequence Diagram

3.3.4 State Transition Diagram

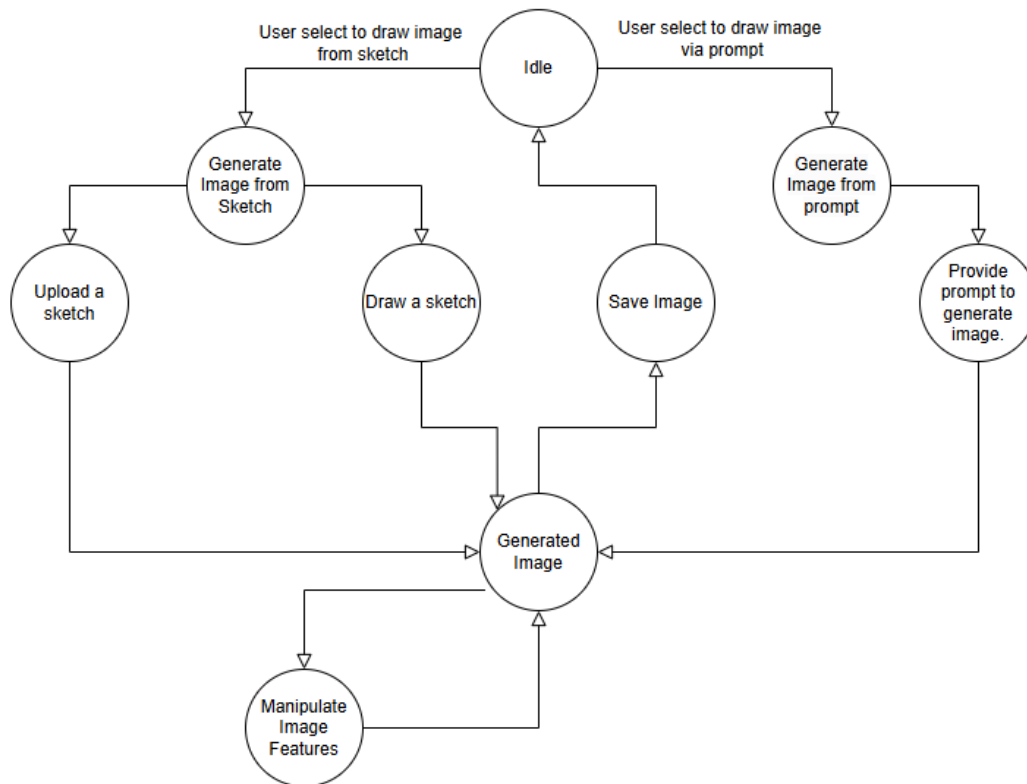


Figure 3.8: State Transition Diagram

3.4 Data Design

This section explains how the major data entities are organized, processed, and stored within the system. In the Sketch2Face, the information domain consists of various inputs (sketches, textual prompts) and outputs (generated and manipulated facial images).

3.4.1 Data Representation Diagram

For the representation of our data, we selected Entity-Relationship Diagram for this purpose which is as following :

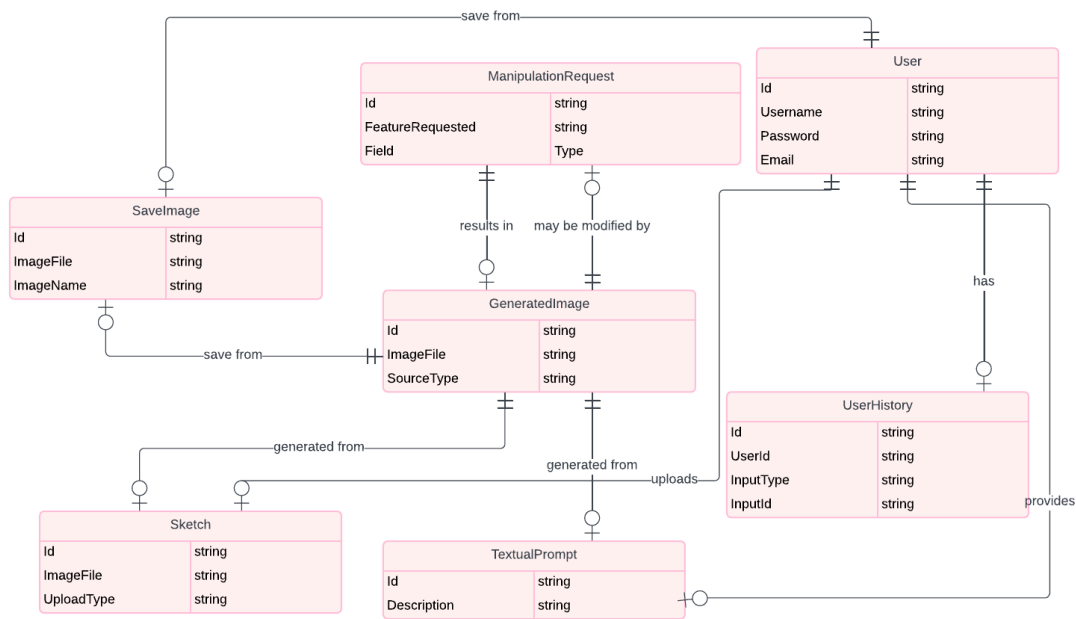


Figure 3.9: ERD Diagram

3.4.2 Data Structures

The primary data structures in Sketch2Face include:

3.4.2.1 Sketch Data

- **Input**: Hand-drawn sketches uploaded or drawn on GUI by the user.
- **Structure**: Stored as image files (e.g., PNG or JPEG) in a temporary storage system before processing.

- **Processing:** The sketches are passed through an AI model, which converts them into a facial image.
- **Storage:** The processed sketch is saved temporarily for manipulation or can be saved directly in database if no manipulation is required.

3.4.2.2 Textual Prompt Data

- **Input:** Textual descriptions provided by the user (e.g., "male, short hair, blue eyes").
- **Structure:** The data/text is passed through AI model for image generation.
- **Processing:** Each attribute (hair color, facial features, etc.) is mapped to specific facial characteristics during the image generation process.
- **Storage:** Once converted to an image, it can be further manipulated or stored directly into database.

3.4.2.3 Generated Image Data

- **Output:** The AI model generates facial images from the user's sketches or textual prompts.
- **Structure:** Stored as image files (e.g., PNG, JPEG).
- **Storage:** Image can be further manipulated or stored directly into database.

3.4.2.4 Manipulation Data

- **Input:** User requests to manipulate specific features of the generated image (e.g., change hair color or adjust facial shape).
- **Processing:** The AI model updates the image based on the modifications, and the new image is stored in the system.
- **Output:** Modified image.
- **Storage:** Image can be further manipulated or stored directly into database if user is satisfied with modification.

3.4.2.5 User Data

- **Input:** User profiles include login information, history of inputs (sketches, text prompts), and saved images.
- **Structure:** Stored in a relational database with tables for user details, login credentials, and historical actions.

3.4.3 Databases and Data Storage

The system utilizes a combination of file storage and database management systems to handle large amounts of image data and user information.

3.4.3.1 Database (Relational Database)

- **User Profiles:** A relational database that stores user information, including login details, history of actions, and saved images.
- **Sketch/Prompt History:** Tracks sketches and text prompts submitted by the user, mapping each submission to the respective generated image.
- **Manipulation History:** Stores details of the changes made to the images (e.g., color, facial structure modifications).

3.4.4 Organization and Processing

3.4.4.1 Data Storage

- All generated and manipulated images are stored with unique identifiers that map them to the original input (sketch or text prompt).

3.4.4.2 Data Processing

- Sketches and textual prompts are processed in real-time to generate facial images.
- Manipulation requests are processed by applying changes to the image and storing the new version in the database, without overwriting the original.

3.4.4.3 Data Security

- User data (login details, image history) is protected by enforcing strong password to prevent from unauthorized access.
- User can only access their own generated and manipulated images through a secure login process.

Bibliography

- [1] Wikipedia contributors. Data-flow diagram. *Wikipedia, The Free Encyclopedia*, 2024.
https://en.wikipedia.org/wiki/Data-flow_diagram.
- [2] Chang Wook Seo Soyeon Yoon Seongcheol Kim Soohyun Ji Amirsaman Ashtari Jun-yong Noh Kwan Yun, Kwanggyoon Seo. Stylized face sketch extraction via generative prior with limited data. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2024.
- [3] Noberto Rodriguez. Domain modeling: What you need to know before coding. *ThoughtWorks*, 2021.
- [4] Jing-Hao Xue Baoyuan Wu Weihao Xia, Yujiu Yang. Tedigan: Text-guided diverse face image generation and manipulation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2021.
- [4]
- [2]
- [1]
- [3]